
Mechanistic Attribution of Relational Transformer Behavior to Synthetic Pretraining Generators

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Abstract

1 When a relational foundation model (RFM) is pretrained on synthetic relational
2 databases, benchmark comparisons tell us *which* generator produced a better model
3 but not *why*, or whether the advantage reflects a genuine relational skill rather
4 than better marginal statistics. We treat this as a mechanistic data-attribution problem:
5 given four Relational Transformer (RT) checkpoints that share architecture,
6 initialization, and step budget and differ only in the synthetic-data generator that
7 produced their pretraining corpus (RelDiff, GRDM, PluRel, RDB-PFN), we trace
8 the generator’s fingerprint from weight space, through learned representations and
9 attention structure, to the model’s downstream benchmark behavior. Weight-space
10 divergence, representational similarity (CKA), and within-neighborhood attention
11 entropy all agree on the same ranking: only the generator whose synthetic construction
12 makes relational structure necessary for reconstructing masked values (RelDiff)
13 yields a model that mechanistically relies on cross-table structure out of
14 distribution. We then close the loop causally: intervening on exactly the mechanism
15 identified by these diagnostics — corrupting foreign-key structure or zeroing the
16 cross-table attention stream at inference — reproduces the model’s downstream
17 degradation quantitatively, including on the specific tasks and context lengths where
18 the mechanism predicts an effect and where it does not. This provides a template
19 for attributing foundation-model capabilities to specific, checkable properties of a
20 synthetic pretraining corpus, rather than to the generator’s identity alone.

21 1 Introduction

22 A growing family of relational foundation models (RFMs) — Griffin [5], KumoRFM [6], and the
23 Relational Transformer (RT) [7] among them — are pretrained on large corpora of synthetic relational
24 databases rather than on a single fixed dataset, because real multi-table databases [1, 2] with the
25 necessary scale and diversity are scarce and often proprietary. This creates a data-attribution problem
26 that is one step removed from the usual training-example-level setting studied at this workshop: the
27 object whose contribution we want to attribute is not a training example but a *generator* — a procedure
28 that induces a distribution over synthetic databases. Downstream benchmarks can rank generators, but
29 a benchmark ranking is a black-box comparison: it does not say what property of a generator’s output
30 caused the ranking, whether the advantage will transfer to schemas unlike anything the generator
31 produced, or whether the model actually learned to *use* relational structure versus merely fitting
32 the generator’s marginal statistics. Because the pretraining objective is masked-cell reconstruction,
33 this last question is exactly the ambiguity mechanistic interpretability is suited to resolve: a well-fit
34 marginal distribution and a genuine relational computation both lower reconstruction loss, but only
35 one of them is a transferable skill.

36 We study this question in the cleanest setting available: four RT checkpoints, byte-identical in
37 architecture (12 blocks, $d=256$, 19.2M parameters, three attention streams per block), initialization,

and step budget, pretrained on corpora from four synthetic relational-database generators representing distinct design philosophies — two condition on real reference databases (RelDiff: joint multi-hop graph diffusion; GRDM: graph-conditional diffusion with 1-hop locality) and two synthesize schemas and values from scratch with no reference data (PluRel, RDB-PFN). Because the only experimental variable across the four arms is the generator, any measured difference between the four checkpoints is attributable to the data it produced, not to confounds of architecture or optimization.

Contributions. (1) We measure, in the corpora themselves and with no model involved, how much of a masked cell’s recoverable signal lives in neighbouring tables (Sec. 3); this turns the property our account rests on into a measured quantity that already orders the generators. (2) We give a weight-space and representational diagnostic (Sec. 4) that predicts, from a checkpoint alone, whether a generator taught the model to treat cross-table structure as predictive or to pool over it. (3) We show this diagnostic is causally load-bearing, not merely correlational (Sec. 5): ablating the identified mechanism at inference reproduces the corresponding benchmark degradation, including a case where the mechanism’s own failure mode — misleading structural priors on one schema — correctly predicts a benchmark loss rather than a gain. Together these give a recipe for attributing what a synthetic-data generator teaches a foundation model beyond a single benchmark number.

2 Setup: a controlled comparison across four generators

Generators. The four generators span the two structurally distinct families of reference-free vs. reference-conditioned synthesis. *RelDiff* [8] and *GRDM* [10] copy a real reference database’s schema and foreign-key (FK) graph and regenerate attributes with a diffusion model, jointly (RelDiff, multi-hop) or locally (GRDM, 1-hop, non-diffusion in-degree sampling for structure). *PluRel* [9] and *RDB-PFN* [11] require no reference: they sample a schema and PK-FK graph from a hand-designed prior and generate attribute values with structural causal models. Each arm was pretrained on a corpus of 1,000 generated databases. None of the four learns a schema prior from data — schema is either copied verbatim or hand-designed — so any difference in downstream relational behavior must come from how each generator couples attribute values to graph structure, not from schema diversity per se.

Models. Each of the four RT checkpoints (`reldiff_final.pt`, `grdm_final.pt`, `plurel_final.pt`, `rdpfn_final.pt`) was pretrained with the identical masked-value-reconstruction objective, encoder set (typed value encoders for number/text/datetime/boolean, 384-d MiniLM [13] text embeddings), and step budget, from a shared initialization. The shared initialization is verifiable from the checkpoints rather than assumed: 132 tensors (115,585 parameters, chiefly decoder heads and type norms) are *bitwise* identical across all four arms, which independently trained models cannot achieve by coincidence. The RT architecture separates attention into three interpretable streams per block: `col` (column-level), `feat` (row/feature-level, which includes a cell’s parent rows via `kv_in_f2p`), and `nbr` (neighbor attention, architecturally masked to FK-linked cells). Both cross-table routes are built from the same `f2p_nbr_idx` tensor, and intervening on each separately shows they operate *in series* rather than in parallel: `nbr` lets a parent aggregate its children, but that aggregate reaches a masked cell only through `feat`’s parent-cell inclusion. Blocking the `feat` route costs RelDiff +0.193 masked-cell loss and additionally zeroing `nbr` costs nothing further (+0.193, identical), while zeroing `nbr` alone costs +0.132. The `feat` path is therefore the bottleneck through which cross-table information must pass, and `nbr` feeds it rather than substituting for it. This separation is what makes the checkpoints diagnosable: a generator’s imprint on *relational* computation and its imprint on *marginal/attribute* statistics land in different, nameable parts of the network.

Evaluation. All mechanistic probes (Sec. 4) use three RelBench databases (`rel-ratebeer`, `rel-arxiv`, `rel-stack`) that lie outside every generator’s pretraining reference set, so results reflect out-of-distribution transfer rather than memorization of a shared reference. The downstream benchmark (Sec. 5) covers 9 RelBench tasks across 5 in-context lengths (100–30,000 tokens), the same in-context-learning protocol used to compare the checkpoints as generative pretrained models.

3 What the corpora make predictable

The account above rests on *predictive necessity*: because the pretraining objective is masked-cell reconstruction, a corpus can only teach cross-table computation if its masked cells are hard to recover

Table 1: Neighbour predictive information per corpus, measured with no RT involved (40 databases each; median, 95% bootstrap CI). RelDiff is the only corpus in which reading a neighbouring table measurably helps recover a held-out cell. RDB-PFN is its mirror image: the most learnable within-row structure of any corpus and no relational signal at all.

Corpus	Tasks	Own-row R^2	Parent lift	Child lift
RelDiff	48	0.022	+0.072 [+0.018, +0.091]	−0.001
PluRel	113	0.166	+0.009 [+0.001, +0.022]	−0.000
RDB-PFN	120	0.436	+0.001 [−0.004, +0.009]	−0.005
GRDM	120	−0.011	−0.057 [−0.076, −0.044]	−0.105

90 from their own row. That is a property of the data, not of any model, so we measure it directly and
91 without involving the RT at all.

92 For each corpus we sample 40 databases, recover the FK graph, and for each target column fit
93 identical gradient-boosted regressors three times: from the target’s own row; from its own row plus
94 the columns of its FK parents; and from its own row plus aggregates over its FK children. The
95 gain from adding relational features is the neighbour predictive information in that corpus. Splitting
96 parents from children matters because the two map onto different routes in the RT: a cell reading its
97 parents flows through the *feat* stream’s parent-cell inclusion, while aggregation over children is what
98 the *nbr* stream does. Scores are clipped to $[-1, 1]$ before differencing, since unbounded negative R^2
99 lets one diverged fit dominate a mean, and we report medians with 95% bootstrap intervals.

100 Table 1 orders the generators the same way every downstream measurement does, and it does so
101 before any model is trained. RelDiff’s parent lift is eight times PluRel’s and roughly seventy times
102 RDB-PFN’s, whose interval spans zero. Read as a share of the predictable signal, neighbours
103 contribute 77% for RelDiff, 5% for PluRel and 0.1% for RDB-PFN. RelDiff’s cells are close to
104 unrecoverable from their own row (R^2 0.022) and become recoverable once a parent is visible, which
105 is precisely the condition under which masked-cell reconstruction rewards learning to read across
106 tables. RDB-PFN’s profile — the most predictable rows and no relational signal — is the data-side
107 counterpart of its benchmark behaviour, strong at short context and untouched by ablation.

108 Two qualifications. Child lift is flat in every corpus, RelDiff included, so what its data rewards
109 specifically is reading parents; the transfer to child-aggregation tasks at benchmark time is not
110 something the corpus directly incentivises, and our aggregates (mean, standard deviation, count) may
111 simply be too coarse to capture it. Second, GRDM’s negative lift is best read alongside a property
112 of its shipped corpus rather than its design: every string and datetime column was converted to a
113 float and never converted back, so its databases carry no text or categorical columns at all (0 against
114 RelDiff’s 234 over the same references). The RT’s text encoder and MiniLM pathway therefore
115 received no signal in that arm, which is a confound worth stating plainly: the GRDM arm differs
116 from the others in more than its generator.

117 **Diversity is not the mechanism.** A natural alternative is that RelDiff simply produces more diverse
118 data. Measuring feature, table and cross-table diversity directly from the generated databases —
119 never through any of the four models, so the measurement cannot be an artifact of what it explains —
120 RelDiff is indeed most diverse on all three axes (1.10 / 0.57 / 1.04). But the ranking of the others
121 does not track mechanism formation: PluRel exceeds GRDM on cross-table diversity (0.85 vs. 0.71)
122 yet learns no usable relational computation (Table 4), while GRDM does on in-distribution schemas.
123 Diversity describes a generator’s marginal output; predictive necessity describes whether values are
124 *coupled* to structure, and only the latter separates the four models.

125 4 Mechanistic fingerprint: weight space, function, and attention

126 **Weight space (E1).** Because all four arms share an initialization, we measure each checkpoint’s
127 deviation from the 4-model mean, grouped by stream and by depth. Table 2 shows the deviation
128 norm for each attention stream. RelDiff is the only arm whose largest deviation is in *nbr*-attention
129 (6.43, vs. 3.8–5.6 for the other three, where *nbr* is always the *smallest* deviating stream); GRDM’s
130 largest deviation is in *feat*-attention (7.18). PluRel and RDB-PFN move least overall and in a shared
131 direction (pairwise deviation cosine +0.11, against a $-1/3$ baseline for unrelated directions). This

Table 2: E1: per-stream weight deviation from the 4-model mean (L2 norm; larger = more specialized by that stream). RelDiff uniquely peaks in nbr; GRDM peaks in feat; PluRel/RDB-PFN are near-identical and diffuse.

Model	Encoders	Col-attn	Feat-attn	Nbr-attn	FFN
GRDM	1.95	6.10	7.18	5.55	10.82
PluRel	1.05	5.13	4.68	3.90	7.69
RDB-PFN	1.03	5.57	4.35	3.80	7.51
RelDiff	1.64	5.37	5.88	6.43	9.49

Table 3: E4 (CKA, real data) and E5 (within-FK-neighborhood attention). Real-reference generators (RelDiff, GRDM) produce the most functionally divergent-yet-mutually-similar late representations and the sharpest (lowest-entropy) attention within the FK neighborhood; PluRel/RDB-PFN pool near-uniformly.

E4: CKA (early / late blocks)			E5: attention within FK neighborhood		
Pair	Early	Late	Model	Norm. entropy	Top-1 mass
GRDM~RelDiff	0.944	0.824	RelDiff	0.843	0.396
PluRel~RelDiff	0.944	0.810	GRDM	0.858	0.394
GRDM~PluRel	0.938	0.718	RDB-PFN	0.890	0.361
PluRel~RDB-PFN	0.909	0.707	PluRel	0.898	0.361
RDB-PFN~RelDiff	0.870	0.587			
GRDM~RDB-PFN	0.808	0.406			

132 is a purely weight-based signature, available in minutes on a CPU without any dataloader, and it
 133 predicts the functional and behavioral results below *before* they were measured.

134 **Function (E4).** Weight distance can overstate differences if different weights compute the same
 135 function, so we also measure linear CKA between residual-stream activations on identical real-data
 136 batches (Table 3, left/right = early/late blocks). Functional divergence grows monotonically with
 137 depth for every pair: all four models agree on early featurization ($\text{CKA} \geq 0.80$ in blocks 0–5) and
 138 diverge in late relational integration. Notably, function and weights disagree on which models cluster:
 139 GRDM and RelDiff — weight-space outliers in Table 2 — are the most functionally *similar* pair
 140 late (0.824), while RDB-PFN, PluRel’s weight-space twin, is the functional outlier (as low as 0.406
 141 against GRDM). Weight direction measures what a corpus pushed on; CKA measures where the
 142 computation landed on real inputs; both are needed because they can disagree.

143 **Attention (E5).** The RT’s nbr attention is architecturally masked to FK-linked cells (≤ 5 edges/row),
 144 so whether heads can attend across tables is fixed by construction; what a generator can teach is *how*
 145 *attention is distributed within* that permitted neighborhood. We hook wq/wk inputs and recompute
 146 masked softmax scores on fixed probe batches. Table 3 (right) shows RelDiff and GRDM — the
 147 two real-reference generators — produce measurably sharper (lower-entropy) within-neighborhood
 148 attention than PluRel and RDB-PFN, which are close to uniform. Selection versus pooling is exactly
 149 the operation that should behave differently as context grows: pooling dilutes with more neighbors,
 150 selection does not.

151 **Behavioral reliance (E7).** Attention structure shows what information *could* be used; we test what is
 152 *actually used* with input corruptions on real held-out data: intact inputs, FK links shuffled within a
 153 table (structure destroyed, marginals intact), and neighbors dropped. We define a structure-reliance
 154 index as the reconstruction-loss increase under FK shuffling, and separately measure the loss increase
 155 from zeroing the nbr stream at inference. Table 4 shows RelDiff is the only model with non-trivial
 156 reliance on either intervention (+20% loss under FK shuffling); the other three are within $\pm 2\%$
 157 of intact loss under every corruption, i.e., functionally indifferent to relational structure on unseen
 158 schemas, despite GRDM’s sharp attention in Table 3. Blanking the FK links rather than shuffling
 159 them reproduces the same pattern (RelDiff +0.065, all others ≤ 0.014), so the effect is not an artifact
 160 of the particular corruption used. Two controls sharpen this. Corrupting a graded fraction of FK
 161 links rather than all of them yields a monotone loss curve for RelDiff and near-flat lines for the rest,
 162 with RelDiff’s slope 11–29 $\times$ the others’. And an *untrained* RT, scored identically, already degrades
 163 4.9% under FK shuffling, simply because altering the mask alters the computation. Against that floor
 164 the three structure-blind arms sit at 0.7–1.3 $\times$ — statistically indistinguishable from a network that

Table 4: E7: masked-reconstruction loss under corruption (mean over 3 held-out real DBs, 27 probe batches). RelDiff is the only model that degrades under structural corruption; all others are within noise.

Model	Intact loss	Δ nbr zeroed	Δ FK shuffled	Δ FK blanked
RelDiff	0.339	+0.029	+0.069 (+20%)	+0.065
PluRel	0.339	+0.007	+0.009	+0.014
GRDM	0.350	+0.013	−0.005	−0.007
RDB-PFN	0.351	−0.002	+0.003	+0.007

never trained — while RelDiff reaches $17\times$. Their indifference is therefore not a small effect but the absence of one.¹ GRDM’s relational knowledge is *schema-specific*: it learned regularities of its own reference databases rather than a transferable use of FK structure.

The three diagnostics agree: RelDiff’s corpus is the only one that makes cross-table structure predictively necessary for reconstruction, and that is the only corpus whose model both attends selectively within the FK neighborhood and functionally relies on it out of distribution. This is consistent with the underlying generative process — RelDiff regenerates attributes *conditioned on* neighboring tables via joint multi-hop diffusion, so a masked value’s neighbors genuinely carry information about it, and training pressure lands on the nbr stream (E1) before we ever measure function or behavior.

5 Closing the loop: does the mechanism cause the benchmark result?

Sec. 4 establishes a correlational story: the generator that induces relational reliance also produces the model with the best downstream benchmark profile (mean AUROC over the six classification tasks at context 1024: RelDiff 0.748 > RDB-PFN 0.740 > PluRel 0.723 > GRDM 0.666), and RelDiff degrades least as context grows from 1024 to 30,000 tokens. This comparison needed care, because the two tasks that drive it are rare-label ones (`rel-stack` user-engagement and user-badge) that at our original $n=256$ contributed only six and ten positive examples, enough for a Hanley–McNeil interval [14] of ± 0.073 . We therefore rescored both tasks at 30k context with $n=2048$ (≈ 60 positives each). The re-scored numbers move exactly as small-sample regression predicts: RelDiff’s inflated estimates fall (user-engagement $0.929 \rightarrow 0.865$, user-badge $0.847 \rightarrow 0.812$) while the pooling models’ deflated ones rise (PluRel $0.564 \rightarrow 0.652$, RDB-PFN $0.538 \rightarrow 0.642$ on user-engagement), and the already-well-powered cells move by at most 0.007. Corrected, *every* model degrades with context and RelDiff simply degrades least, by a factor of three or more: -0.011 against -0.038 (GRDM), -0.043 (PluRel) and -0.069 (RDB-PFN). Its advantage on the two tasks themselves is undiminished and large (user-engagement 0.865 against 0.652 for the next best). We therefore rely on the ordering, which is robust, and explicitly do not claim that any model improves with context. Correlation between per-(model, database) structure-reliance (Table 4) and context-scaling behavior is 0.71 across all 12 cells (0.82 at the generator level, $n=4$).

Correlation is not causation, so we rerun the exact benchmark protocol under the same mechanistic interventions used in Sec. 4 — FK links shuffled, or the nbr stream zeroed at inference — and pre-registered four predictions before running the ablated benchmark: (1) RelDiff’s advantage on relational, long-context tasks collapses toward the pooling models’ level; (2) its 30k-context advantage specifically disappears; (3) PluRel/RDB-PFN are ablation-indifferent; (4) on `rel-arxiv`, where E7 found a sign-flipped reliance (-0.112 : shuffling FK links *improves* RelDiff), ablation should hurt least or help.

Table 5 confirms predictions (1)–(3). FK corruption erases RelDiff’s engagement advantage at both context lengths, and at 30k it costs 0.286 AUROC — dropping RelDiff from 0.865 to 0.579, below the 0.642–0.652 the structure-blind models reach with no intervention at all. Long context without the relational mechanism is not merely less useful; it is worse than a shorter context, because pooling over more neighbours adds variance and no signal. PluRel and RDB-PFN are unmoved by either intervention at either length ($|\Delta| \leq 0.017$), which is what a within-row computation predicts. Prediction (4) is refuted in its specific location — `rel-arxiv` paper-citation shows only

¹The graded and untrained-floor controls were run on `rel-stack` alone rather than the full three-database probe suite, so their absolute losses are not comparable to the table; the ratios are what we report.

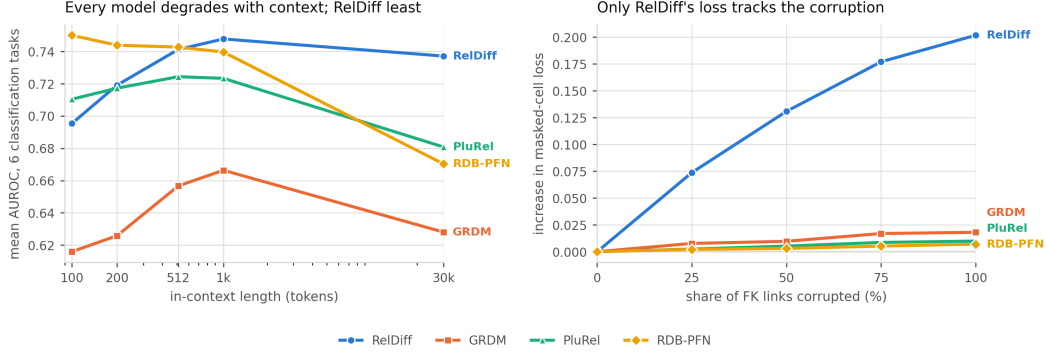


Figure 1: Left: mean AUROC over the six classification tasks against in-context length. The 30k points for the two `rel-stack` tasks are our $n=2048$ rescoring; at the original $n=256$ they carried only six and ten positive examples. Every model degrades with context and RelDiff degrades least, by a factor of three or more. Right: masked-cell loss as a graded fraction of FK links is corrupted. RelDiff’s loss tracks the corruption monotonically at $11\text{--}29\times$ the slope of the others, which are flat.

Table 5: X1: the exact downstream benchmark, rerun under mechanistic ablation, on RelDiff’s largest win (`rel-stack` user-engagement). Ablating exactly the mechanism identified by E1/E5/E7 collapses RelDiff to the pooling models’ level; PluRel/RDB-PFN are unaffected by either intervention. All rows use $n=2048$ (≈ 60 positives per cell); the 30k rows were rescored at this sample size after we found the original $n=256$ estimates unstable. RelDiff is the only model that moves: corrupting FK structure costs it 0.286 AUROC at 30k, dropping it below the level the structure-blind models reach untouched. PluRel and RDB-PFN are indifferent to both interventions everywhere ($|\Delta| \leq 0.017$), as a bag-of-rows computation should be. GRDM is the informative exception: FK corruption *improves* it by 0.102, the misleading-priors effect that also appears in E7’s per-database signs.

Ctx	Model	Intact	FK shuffled	Nbr zeroed	Δ_{FK}	Δ_{nbr}
30,000	RelDiff	0.865	0.579	0.624	−0.286	−0.241
30,000	PluRel	0.652	0.650	0.636	−0.003	−0.017
30,000	RDB-PFN	0.642	0.643	0.634	+0.001	−0.008
30,000	GRDM	0.520	0.622	0.537	+0.102	+0.018
1,024	RelDiff	0.847	0.718	0.728	−0.129	−0.119
1,024	PluRel	0.708	0.701	0.697	−0.007	−0.011
1,024	RDB-PFN	0.691	0.694	0.687	+0.003	−0.004
1,024	GRDM	0.609	0.618	0.622	+0.009	+0.013

207 mild, ordinary reliance — but the phenomenon it anticipated appears twice elsewhere, and in the
208 stronger form: on `rel-ratebeer` user-churn FK-shuffling improves both structure-using models
209 (+0.061, +0.064), and on `rel-stack` user-engagement at 30k it improves GRDM by +0.102.
210 Where a model’s relational priors fit the task the mechanism produces its wins; where they misfit, the
211 same mechanism produces its losses, and removing it helps.

212 Taken together, this is a causal double confirmation: the pretraining corpus determines a checkable
213 mechanism (E1/E4/E5/E7), and surgically removing that mechanism — and only that mechanism,
214 holding the model otherwise fixed — reproduces the corresponding change in downstream capability,
215 including its failure mode. The generator did not just correlate with a better benchmark score; it built
216 the specific computation the benchmark score depends on.

217 6 Related work

218 RFM architectures differ in how they model cross-table interaction: message passing [1, 3, 5]
219 vs. relational attention [4, 6, 7]. We build on the RT because its factored attention streams make
220 attribution to a specific computational channel possible, which a message-passing update does not as
221 directly expose. Generators split into reference-conditioned families that regenerate a real schema
222 and FK graph [8, 10] and reference-free families that sample both from scratch [9, 11]. Prior

comparisons rank them by benchmark transfer; we take that ranking as a starting point and ask what mechanism produces it, using linear CKA [12] for functional rather than weight-level comparison and input/activation ablation for causal validation.

7 Discussion and limitations

Our central claim is methodological: benchmark rankings of synthetic-data generators can be decomposed into a checkable mechanism, and that mechanism can be causally validated by ablating exactly what the diagnostics point to, rather than by ablating the generator’s identity (retraining from scratch). This is attractive for practitioners choosing or designing a generator for RFM pretraining, because the weight-space diagnostic (E1) is available in minutes on a CPU, before any expensive downstream evaluation, and the ablation protocol (E7/X1) reuses the pretraining objective rather than requiring a new benchmark.

The main limitations are scope and scale. We study one architecture (RT) and one pretraining objective (masked-cell reconstruction); the extent to which weight-stream separation transfers to message-passing RFMs is untested. Each generator arm is a single pretraining run at a short (3.5-hour) step budget shared across arms, so we report the *direction* of weight-space specialization rather than claiming the absolute magnitudes would hold at production-scale pretraining; the CKA and behavioral results (Sec. 4–5), being measured directly on the trained checkpoints, are not subject to this caveat. Finally, reference-conditioning is neither necessary nor sufficient in isolation: GRDM uses real references yet its relational computation does not transfer beyond its reference schemas. What matters is whether a generator’s construction makes cross-table structure predictively necessary (Sec. 3), which a reference-free generator could in principle also achieve.

8 Conclusion

Benchmark comparisons of synthetic-data generators answer *which* is better, not *why*. We show the question is answerable end to end: the property that matters is measurable in the corpus before any model exists, it leaves a predictable imprint in weight space, and ablating the mechanism it builds removes the benchmark advantage it explains. We hope this recipe — corpus measurement, checkpoint diagnostics, causal ablation — generalizes to attributing other foundation-model capabilities to checkable properties of their pretraining data.

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281 A Supporting figures

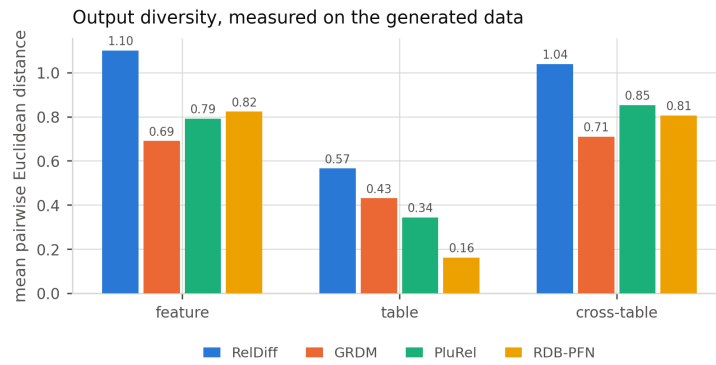


Figure 2: Output diversity of each generator, computed directly from the generated databases rather than through any model’s representation. RelDiff leads on all three axes, but the ranking of the other three does not track mechanism formation: PluRel exceeds GRDM on cross-table diversity yet learns no usable relational computation.

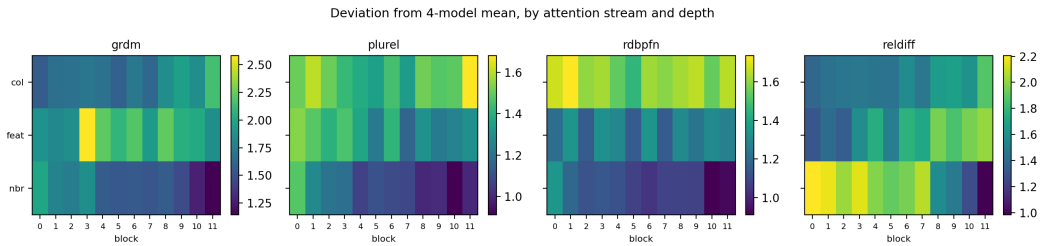


Figure 3: E1: deviation from the four-model mean by attention stream and depth. RelDiff is the only arm whose deviation peaks in nbr; GRDM’s peaks in feat.

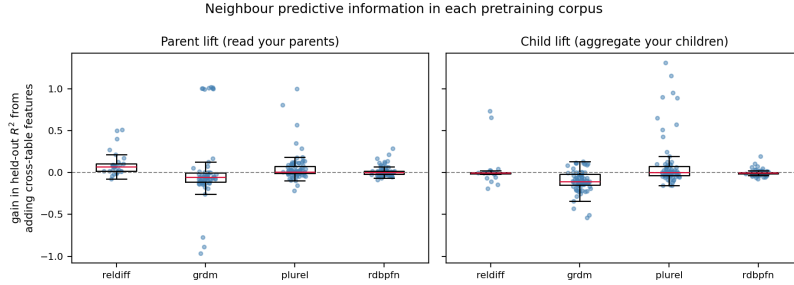


Figure 4: Neighbour predictive information per corpus (Sec. 3). Each point is one target column; the gain is from adding cross-table features to a within-row baseline.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and Sec. 1 state exactly the three contributions reported in Sec. 3–5: a model-free measurement of neighbour predictive information in the corpora, a weight-space diagnostic, and its causal validation via ablation; we do not claim generalization beyond the single architecture and objective studied (see Limitations).

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 7 (Discussion and limitations) discusses the single-architecture/single-objective scope, the single pretraining run per generator arm at a short shared step budget, and that reference-conditioning is neither necessary nor sufficient for mechanism formation.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper is purely empirical/mechanistic; it contains no theorems or formal proofs.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Sec. 2 specifies the shared architecture, initialization, step budget, and objective across the four checkpoints; Sec. 4–5 specify the exact probe/ablation protocols (corruption types, probe databases, context lengths, and the causal-ablation predictions registered before running X1) needed to reproduce each table.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: The four pretraining corpora, checkpoints, and analysis code are being prepared for a public release accompanying the camera-ready version and are not yet hosted at an anonymized URL at submission time.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Sec. 2 specifies the architecture, encoders, and shared pretraining protocol; Sec. 4 specifies the held-out, out-of-reference-set evaluation databases and probe-batch protocol used for every mechanistic measurement.

7. Experiment statistical significance

Answer: [No]

Justification: Each generator arm corresponds to a single pretraining run (shared compute budget across four arms), so we do not report error bars across seeds; we mitigate this by cross-validating each finding across four independent diagnostics (E1, E4, E5, E7) and a causal ablation (X1) rather than relying on any single point estimate, and we discuss this as a limitation in Sec. 7.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Sec. 4 notes that the weight-space diagnostic (E1) runs in minutes on a single CPU with no dataloader, while the representational, attention, and ablation probes (E4, E5, E7) and the causal-bridge benchmark (X1) require a GPU box for activation extraction and inference; Sec. 7 states the shared 3.5-hour pretraining budget per arm.

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

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10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Sec. 7 frames the positive impact (a cheap, pre-deployment diagnostic for whether a synthetic-data pipeline taught a foundation model a genuine, transferable skill rather than a benchmark-specific artifact); we are not aware of a direct negative-impact pathway specific to this attribution methodology beyond those already inherent to relational foundation models generally.

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse?

Answer: [N/A]

Justification: The released checkpoints are small (19.2M-parameter) research artifacts trained on synthetic and public benchmark relational data; they pose no high-risk dual-use profile requiring gated release.

12. Licenses for existing assets

Question: Are the creators or original owners of assets used in the paper properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

369 Justification: We cite the RelBench benchmark and the RT architecture and each of the four
 370 generators (RelDiff, GRDM, PluRel, RDB-PFN) by their source papers/repositories in the
 371 References, and use each only under its stated license and terms.

372 **13. New assets**

373 Question: Are new assets introduced in the paper well documented and is the documentation
 374 provided alongside the assets?

375 Answer: [\[Yes\]](#)

376 Justification: The four pretrained RT checkpoints and the mechanistic-analysis probes
 377 (E1–E7, X1–X2) are new artifacts of this work; Sec. 2–5 document their construction and
 378 intended use, and full documentation will accompany the code/checkpoint release referenced
 379 in item 5.

380 **14. Crowdsourcing and research with human subjects**

381 Answer: [\[N/A\]](#)

382 Justification: The paper does not involve crowdsourcing or human subjects.

383 **15. Institutional review board (IRB) approvals or equivalent for research with human**
 384 **subjects**

385 Answer: [\[N/A\]](#)

386 Justification: The paper does not involve human subjects research.

387 **16. Declaration of LLM usage**

388 Question: Does the paper describe the usage of LLMs if it is an important, original, or
 389 non-standard component of the core methods in this research?

390 Answer: [\[N/A\]](#)

391 Justification: The only LLM-adjacent component is a fixed, off-the-shelf MiniLM text
 392 encoder inherited unmodified from the pre-existing RT architecture (Sec. 2); it is not a novel
 393 or original component of this paper’s core method, which is the mechanistic attribution
 394 analysis itself.